

The missing pressure traction in Scott–Vogelius–Nitsche methods on polygonal approximations of a curved wall

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Abstract

We consider two-dimensional Stokes flow in which a curved no-slip wall is replaced by an inscribed polygon with edges of length at most h_Γ . Following Gjerde and Scott, the flow is discretised on a mesh of size h_Ω by exactly divergence-free Scott–Vogelius elements of degree $k \geq 4$, with no-slip imposed on the polygon by Nitsche’s method with penalty μ/h_Ω . Scott asked whether the H^1 error is $O(h_\Gamma^{3/2} + h_\Omega^k)$ for general Stokes data, and if not, why not. For the method as printed and a fixed penalty, the answer is no whenever the pressure varies along the wall. The Nitsche form has no pressure traction, so the discrete flow leaks through the polygon with normal velocity $(h_\Omega/\mu)(p - \bar{p})$, where $\bar{p}$ is the mean wall pressure. We prove that this leak is the leading error, with constant 1 in the slip and energy norms, and that the energy and H^1 errors are of order $(h_\Omega/\mu)^{1/2}$ and h_Ω/μ , with lower bounds. A mean-free traction term, whose velocity coincides with that of a zero-net-flux formulation of Frachon, Nilsson and Zahedi transcribed to this setting, attains $h_\Gamma^{3/2} + h_\Omega^k$ in H^1 for general data, and the exponent 3/2 is sharp for every $k \geq 4$. A penalty scaling like $h_\Omega/\min_e |e|$ is sufficient, and necessary under a shape condition on one boundary triangle, by a witness certified in exact arithmetic. Computations with $k = 4$ agree with the predicted rates, constants and threshold.

Keywords: Stokes equations, Scott–Vogelius elements, Nitsche’s method, curved boundaries, penalty parameter

1. Introduction

For $k \geq 4$ on meshes without singular vertices, the divergence maps continuous piecewise- P_k velocities onto discontinuous piecewise- P_{k-1} pressures [1, 2], so Scott–Vogelius velocities are divergence-free pointwise. On fitted meshes with strong no-slip data this gives exact mass conservation and pressure-robust velocity errors [3]. At a curved wall the same property causes trouble. If the wall is replaced by an inscribed polygon and no-slip is imposed strongly, the constraints can force a zero velocity gradient at a polygon vertex (for instance one lying in only two triangles), giving $O(1)$ gradient errors there and an H^1 error of order $h^{1/2}$ [4, 5]. This lock is a two-triangle phenomenon: under (M2) below, which puts every wall vertex in at least three triangles, strong no-slip also attains $h_\Gamma^{3/2} + h^k$ [6]; see Section 4. Gjerde and Scott [5] keep the polygon and impose the wall condition by Nitsche’s method [7], with penalty μ/h_Ω , where h_Ω is the mesh size. Scott observed H^1 errors of order $h^{3/2}$ for a benchmark with zero pressure (since analysed in [8]). He asked for a proof of the error $h_\Gamma^{3/2} + h_\Omega^k$, where h_Γ is the longest polygon edge, for general Stokes data: “If this does not hold, why not?” [4].

The consistent Nitsche form for Stokes flow uses the full traction, so it contains $\langle p, v \cdot n \rangle$ (see, e.g., [9]), a term that is essential for consistency [10]. The form printed in [5, (27)–(28)] omits it. For exactly divergence-free spaces its placement

matters. In both equations it spoils incompressibility near the boundary [11]. In the momentum equation alone [10, 12] it leaves the constant pressure mode free, and Frachon, Nilsson and Zahedi handle this by testing the momentum equation only with zero-net-flux velocities [12, §5.1]. Exactly divergence-free methods of order h^k exist: Liu, Neilan and Otus [13] work inside a curved domain with a boundary multiplier, a zero-flux constraint and a Taylor correction, and isoparametric Piola-mapped Scott–Vogelius elements fit the wall and are pressure-robust [14, Theorem 6.1], also in 3D [15]. Our object is the straight-sided Gjerde–Scott method, for which $h_\Gamma^{3/2}$ is the sharp target.

Without a traction, only the penalty balances the wall pressure. The discrete flow therefore leaks through the polygon with normal velocity $(h_\Omega/\mu)(p - \bar{p})$, where $\bar{p}$ is the mean of p on the wall. This is the consistency error of penalty methods [16], acting on the pressure alone; as $h_\Omega/\mu \rightarrow 0$, the method still converges. *Contributions.* (i) We prove that the leak is the leading error, with constant 1 in the slip and energy norms, and give two-sided bounds of order $(h_\Omega/\mu)^{1/2}$ (energy) and h_Ω/μ (H^1) (Theorem 1). The H^1 rate is 1 because incompressibility turns the normal load into a tangential derivative (Lemma 3). So for fixed penalty, Scott’s estimate holds only when p is constant on the wall. (ii) We prove that the mean-free correction $\langle p_h - \bar{p}_\Gamma(p_h), v \cdot n \rangle$ attains $h_\Gamma^{3/2} + h_\Omega^k$ for general data and every $k \geq 4$ (Theorem 3), and that the exponent 3/2 is sharp for every $k \geq 4$ (Remark 1). The device is not new: its velocity is that of the zero-net-flux formulation of [12], transcribed to this setting. The singularity of the naive form and the loss of incompressibil-

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ity of the L_0^2 alternative (Proposition 2) are elementary; compare [12, (5.7)–(5.9)] and [13, Remark 3.2]. (iii) The penalty threshold scales like $\rho = h_\Omega / \min_e |e|$. This is sufficient, and it is necessary under a shape hypothesis, via a single-triangle witness certified in exact arithmetic (Proposition 4). (iv) Computations with $k = 4$ confirm the rates, the leak constant 1 and the penalty scaling (the penalty study requested in the prize). We treat a disk in a square; Clough–Tocher refinements, general domains, Navier–Stokes, 3D and omitted proofs are in [6]. The analysis and computations were carried out with an AI research assistant, as declared at the end.

2. Setting and the three methods

Let $R = (-L, L)^2$, $L > 1$, D the unit disk, $\Omega = R \setminus \bar{D}$, $\Gamma = \partial D$, and let (u, p) be smooth with $-\Delta u + \nabla p = f$, $\operatorname{div} u = 0$ in Ω , $u|_\Gamma = 0$, $u|_{\partial R} = g$. P_h is a convex polygon with vertices on Γ , $\Gamma_h = \partial P_h$ and $\Omega_h = R \setminus P_h \supset \Omega$. On Γ_h and on Γ , n is the unit normal pointing out of Ω_h (resp. Ω); v is the outward normal of R ; $\langle \cdot, \cdot \rangle$ is the $L^2(\Gamma_h)$ product. A triangulation of Ω_h with edges E_Γ on Γ_h has $h = h_\Omega := \max \operatorname{diam} T$, $h_\Gamma := \max_{E_\Gamma} |e|$, $\rho := h / \min_{E_\Gamma} |e|$ and $\gamma := \mu h_\Gamma / h$ (the effective boundary penalty, $\mu/h = \gamma/h_\Gamma$). We assume (M0) shape regularity and $|e| \geq c_0 h_\Gamma$ on E_Γ , (M1) no singular vertices, and (M2) the angle condition of [2, Thm. 1]; $h_\Gamma \ll h$ is allowed. For $k \geq 4$, V_h is the space of continuous piecewise- P_k fields, V_h^R and V_h^0 are its subspaces vanishing on ∂R and on $\partial\Omega_h$, Π_h is the space of discontinuous piecewise- P_{k-1} functions, and $Z_h := \{v \in V_h^R : \operatorname{div} v = 0\}$. Then $\operatorname{div} V_h^R = \Pi_h$, and $(V_h^0, \Pi_h \cap L_0^2)$ has an inf-sup constant β independent of h [2, [6, Lemmas 4.3, 4.5]]. With $Du = \nabla u + (\nabla u)^\top$, $\partial_n u = (\nabla u)n$ and $\bar{p}_\Gamma(q) := |\Gamma_h|^{-1} \int_{\Gamma_h} q$, the Nitsche form

$$N_h(u, v) = \frac{1}{2}(Du, Dv) - \langle \partial_n u, v \rangle - \langle u, \partial_n v \rangle + (\mu/h) \langle u, v \rangle$$

is the symmetric Nitsche form of [5]. All methods seek $u_h \in V_h$ with the outer data on ∂R and $p_h \in \Pi_h$ with $(q, \operatorname{div} u_h) = 0$ for $q \in \Pi_h$ and

$$N_h(u_h, v) - (p_h, \operatorname{div} v) + b_\Gamma(p_h, v) = (\tilde{f}, v) \quad \forall v \in V_h^R. \quad (1)$$

The printed method GS [5, (28)] has $b_\Gamma = 0$ (we read its test space as V_h^R and use the opposite pressure sign; see [6, §2]). The naive form has $b_\Gamma = \langle p_h, v \cdot n \rangle$, the consistent traction term. The mean-free form CNS* has $b_\Gamma = \langle p_h - \bar{p}_\Gamma(p_h), v \cdot n \rangle$. The L_0^2 variant is the naive form with $p_h, q \in \Pi_h \cap L_0^2(\Omega_h)$. The outer data are the P_k interpolant g_I of g , or $\tilde{g}_I := g_I - (m_R/8L^2)x$ with $m_R := \int_{\partial R} g_I \cdot v$, which has zero net flux, like the exact data (cf. [17]). The velocities of GS and CNS* are divergence-free pointwise.

To compare u_h with u on $\Omega_h \supset \Omega$ we extend u as a divergence-free field. Let $\tilde{u} := \operatorname{curl} \tilde{\psi}$ for a smooth extension $\tilde{\psi}$ of the stream function of u , let $\tilde{p}$ and $\tilde{f}$ extend p and f , and let $F := \tilde{f} + \Delta \tilde{u} - \nabla \tilde{p}$, which vanishes on Ω . Set $|v|^2 := \|\nabla v\|^2 + \frac{\mu}{h} \|v\|_{\Gamma_h}^2 + \sum_{e \in E_\Gamma} |e| \|\partial_n v\|_{L^2(e)}^2$ and $G := \|p - \bar{p}\|_{L^2(\Gamma)}$. Let $\varepsilon_h := h^k + \inf |\tilde{u} - z|$, the infimum over divergence-free $z \in V_h$ with the outer data. For bounded γ , $\varepsilon_h \leq Ch^k$ with data $\tilde{g}_I$, and also with g_I if $h_\Gamma \geq h^{2(\sigma_k - k)}$, where $\sigma_k = k + 2$ (k even)

or $k + 1$ (k odd) [6, Lemmas 4.6, 7.32]. Constants C, c do not depend on $h, h_\Gamma, \mu, \gamma, \rho$; C_p may also depend on $\|p - \bar{p}\|_{W^{1,\infty}(\Gamma)}$.

Lemma 1 ([6, Lemma 4.2]). *Let κ be the Korn constant on V_h^R and C_I the inverse trace constant. If $\mu \geq 4\kappa C_I^2 \rho$, which holds when $\gamma \geq \gamma_0 := 4\kappa C_I^2 / c_0$, then $N_h(v, v) \geq c_1 |v|^2$ and $|N_h(w, v)| \leq M |w| |v|$ on V_h^R , with c_1, M independent of h, μ, ρ .*

3. Main results

Lemma 2 (Consistency). *For $v \in V_h^R$, $N_h(\tilde{u}, v) - (\tilde{p}, \operatorname{div} v) + \langle \tilde{p}, v \cdot n \rangle = (\tilde{f}, v) + \mathcal{G}(v)$, where $|\mathcal{G}(v)| \leq C(1 + \gamma^{1/2}) h_\Gamma^{3/2} |v|$ and $\mathcal{G}(v) := \langle (\nabla \tilde{u})^\top n, v \rangle - \langle \tilde{u}, \partial_n v \rangle + \frac{\mu}{h} \langle \tilde{u}, v \rangle - (F, v)$.*

Proof. Green’s formula gives the identity. Chords lie within $h_\Gamma^2/4$ of Γ , so $\tilde{u} = O(h_\Gamma^2)$ on Γ_h and $|(F, v)| \leq Ch_\Gamma^2 \|\nabla v\|$, which bounds the last three terms. On Γ , $(\nabla u)^\top n = n \operatorname{div} u = 0$. On a chord e the normal tilts by an angle $s + O(h_\Gamma^2)$, where s is the arclength from the midpoint m_e , so $(\nabla \tilde{u})^\top n = s a_e + O(h_\Gamma^2)$ with a_e constant on e . The remainder contributes $Ch_\Gamma^2 \|v\|_{L^1(\Gamma_h)}$. The leading term is $O(h_\Gamma)$ pointwise but odd in s : $|\int_e s a_e \cdot (v - v(m_e))| \leq |e|^2 \|\partial_\tau v\|_{L^1(e)}$, and an inverse trace inequality gives $Ch_\Gamma^{3/2} \|\nabla v\|$. $\square$

For $v \in Z_h$, $\int_{\Gamma_h} v \cdot n = \int_{\Omega_h} \operatorname{div} v = 0$, so for GS

$$N_h(\tilde{u} - u_h, v) = \mathcal{R}(v) + \mathcal{G}(v), \quad \mathcal{R}(v) := -\langle \tilde{p} - \bar{p}, v \cdot n \rangle \quad \forall v \in Z_h. \quad (2)$$

To identify the response to the missing load $\mathcal{R}$, let $\varphi := \chi(r) \int_0^\theta (p - \bar{p}) d\theta'$ in polar coordinates, with a cut-off χ equal to 1 near $r = 1$ and to 0 near ∂R ; then $w := \operatorname{curl} \varphi = -(p - \bar{p})n$ on Γ . With $I_h^{C^1}$ the C^1 piecewise- P_{k+1} interpolant of [18], $v_\varphi := \operatorname{curl} I_h^{C^1} \varphi \in Z_h$, $\|v_\varphi - w\|_{W^{1,\infty}} \leq Ch^k$, and $v_\varphi = -(p - \bar{p})(x^*)n + O(h_\Gamma)$ on Γ_h , where x^* is the radial projection of x onto Γ .

Theorem 1 (The printed method [6, §6]). *Let $\gamma \geq \gamma_0$ and $Y := (1 + \gamma^{1/2}) h_\Gamma^{3/2} + \varepsilon_h$. (a) $\|\tilde{u} - u_h\| \leq C[(h/\mu)^{1/2} G + Y]$. If $G > 0$ and $h \leq h_0(G)$, with h_0 independent of γ , then $\|\tilde{u} - u_h\| \geq (2M)^{-1} (h/\mu)^{1/2} G - CY$. (b) $\|\nabla(\tilde{u} - u_h)\| \leq C_p h/\mu + CY$. (c) Let $G > 0$, fix γ , and let $h \rightarrow 0$ with ρ bounded and $\varepsilon_h \leq Ch^{3/2}$. Then $\tilde{u} - u_h = (h/\mu)v_\varphi + \vartheta_h$ with $\|\vartheta_h\| = o((h/\mu)^{1/2})$ and $\|\vartheta_h\|_{\Gamma_h} = o(h/\mu)$; hence*

$$\frac{\|\tilde{u} - u_h\|_{L^2(\Gamma_h)}}{(h/\mu)G} \rightarrow 1, \quad \frac{\|\tilde{u} - u_h\|}{(h/\mu)^{1/2}G} \rightarrow 1, \\ c \frac{h}{\mu} G \leq \|\nabla(\tilde{u} - u_h)\| \leq C_p \frac{h}{\mu}.$$

The key to (b) is the following lemma; an inverse estimate would only give $|\langle S, \partial_n v \rangle| \leq Ch_\Gamma^{-1/2} \|\nabla v\|$.

Lemma 3 (Normal-load cancellation). *Let τ_e and n_e be the unit tangent and normal of $e \in E_\Gamma$. Let S be edgewise $W^{1,\infty}$ on Γ_h and continuous at its vertices, with $|S \cdot \tau_e| \leq Ch_\Gamma$ on each e . Then $|\langle S, \partial_n v \rangle| \leq C(\|v\|_{\Gamma_h} + h_\Gamma^{1/2} \|\nabla v\|) \leq C \|\nabla v\|$ for $v \in Z_h$.*

Proof. The tangential part is at most $Ch_\Gamma \|\partial_n v\|_{\Gamma_h} \leq Ch_\Gamma^{1/2} \|\nabla v\|$. On e , $\text{div } v = 0$ gives $\partial_n v \cdot n_e = -\partial_\tau(v \cdot \tau_e)$, so with $\sigma_e := S \cdot n_e$, $\int_e \sigma_e \partial_n v \cdot n_e = \int_e \partial_\tau \sigma_e (v \cdot \tau_e) - [\sigma_e v \cdot \tau_e]_{\partial e}$. The integral is at most $C\|v\|_{L^1(e)}$. At a vertex x shared by e and e' the endpoint terms combine to $v(x) \cdot (\sigma_e \tau_e - \sigma_{e'} \tau_{e'}) = O(|e| + |e'|)|v(x)|$, since S is continuous and the edge directions differ by $O(|e| + |e'|)$. The one-dimensional inverse inequality $\|v\|_{L^\infty(e)} \leq C|e|^{-1/2} \|v\|_{L^2(e)}$ for polynomials gives $\sum_e |e| \|v\|_{L^\infty(e)} \leq C|\Gamma_h|^{1/2} \|v\|_{\Gamma_h}$, and $\|v\|_{\Gamma_h} \leq C\|\nabla v\|$ uniformly in h [6, Lemma 3.2]. $\square$

Proof of Theorem 1. Since the projection $\Gamma_h \rightarrow \Gamma$ has Jacobian $1 + O(h_\Gamma^2)$, $|\mathcal{R}(v)| \leq (G + Ch_\Gamma^2)(h/\mu)^{1/2} \|v\|$. (a) Test (2) with $z - u_h \in Z_h$, z attaining ε_h , for the upper bound. For the lower, $\mathcal{R}(v_\varphi) = G^2 + O(h_\Gamma^2 + h^k)$, $|v_\varphi| \leq (\mu/h)^{1/2}(G + Ch_\Gamma^2 + Ch^k) + C$, and $M[\tilde{u} - u_h] \geq (\mathcal{R}(v_\varphi) - |\mathcal{G}(v_\varphi)|)/|v_\varphi|$. (b) Write $z - u_h = \delta_R + \delta_G$ in Z_h with $N_h(\delta_R, \cdot) = \mathcal{R}$ and $N_h(\delta_G, \cdot) = \mathcal{G} - N_h(\tilde{u} - z, \cdot)$ on Z_h , so that $|\delta_G| \leq CY$. For $r_h := \delta_R - (h/\mu)v_\varphi$ and $v \in Z_h$,

$$N_h(r_h, v) = -\langle (\tilde{p} - \bar{p})n + v_\varphi, v \rangle - \frac{h}{\mu} \left[\frac{1}{2} (Dv_\varphi, Dv) - \langle \partial_n v_\varphi, v \rangle - \langle v_\varphi, \partial_n v \rangle \right].$$

On e , $(\tilde{p} - \bar{p})n + v_\varphi = (p - \bar{p})(m_e^* \tau_e + O(h_\Gamma^2 + h^k))$, with m_e^* the projection of m_e . This is odd in s to leading order, so the first term is at most $C_p(h_\Gamma^{3/2} + h^k) \|\nabla v\|$, as in Lemma 2. Lemma 3 with $S = w$ gives $|\langle v_\varphi, \partial_n v \rangle| \leq (C_p + Ch^k h_\Gamma^{-1/2}) \|\nabla v\|$. Testing with $v = r_h$ gives $|r_h| \leq C_p(h/\mu + h_\Gamma^{3/2}) + C\varepsilon_h$, and (b) follows. (c) Here $\vartheta_h = (\tilde{u} - z) + \delta_G + r_h$, so $|\vartheta_h| \leq C_p h/\mu + CY$, $\|\vartheta_h\|_{\Gamma_h} \leq (h/\mu)^{1/2} |\vartheta_h|$, and $\|v_\varphi\|_{\Gamma_h} = G + O(h_\Gamma^2 + h^k)$. Testing (2) with v_φ , all terms other than $(\mu/h)\langle \tilde{u} - u_h, v_\varphi \rangle$ and $\mathcal{R}(v_\varphi)$ are $o(G^2)$ by (a), (b), so $\|\tilde{u} - u_h\|_{\Gamma_h} \geq c(h/\mu)G$, and a uniform trace inequality gives $\|\tilde{u} - u_h\|_{\Gamma_h} \leq C\|\nabla(\tilde{u} - u_h)\| + Ch^{k+1/2}$. $\square$

Proposition 2 (Naive and L_0^2 forms). (i) *The naive system is square, and $(0, 1)$ lies in the kernel of its matrix.* (ii) *Every solution of the L_0^2 variant has $\text{div } u_h \equiv c$ with $c|\Omega_h| = \int_{\partial R} g_I \cdot v + \int_{\Gamma_h} u_h \cdot n$. It solves the naive system if and only if $c = 0$, and then, for $\gamma \geq \gamma_2$, u_h is the CNS* velocity.*

Proof. (i) $-(1, \text{div } v) + \langle 1, v \cdot n \rangle = 0$ for $v \in V_h^R$. (ii) $\text{div } u_h \in \Pi_h$ is orthogonal to $\Pi_h \cap L_0^2$, hence constant. By (i), the naive pressure form at q equals the CNS* form at $q - \bar{p}_{\Gamma_h}(q)$; if $c = 0$, $(u_h, p_h - \bar{p}_{\Gamma_h}(p_h))$ solves CNS*, which is uniquely solvable (Theorem 3). $\square$

Nothing forces $c = 0$: on test A below ($N = 16$) the L_0^2 variant had $\|\text{div } u_h\| = 4.5 \cdot 10^{-3}$.

Theorem 3 (The mean-free method; [6, Thm. 7.2, Rem. 7.3]). *There is $\gamma_2 \asymp \max(\gamma_0, \beta^{-2})$ such that for $\gamma \geq \gamma_2$: (a) CNS* is uniquely solvable, and its velocity equals that of the naive form tested only with $\{v \in V_h^R : \int_{\Gamma_h} v \cdot n = 0\}$, as in [12]; (b) $|\tilde{u} - u_h| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$; (c) consequently, for $\gamma \leq \gamma_{\max}$ and data $\tilde{g}_I$, $\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h_\Omega^k)$, with C depending on $\gamma_{\max}$ (also with g_I under the condition above).*

Sketch. Let $B^*(q, v) := -(q, \text{div } v) + \langle q - \bar{p}_{\Gamma_h}(q), v \cdot n \rangle$. Then $B^*(q + c, v) = B^*(q, v) - c \int_{\Gamma_h} v \cdot n$, and Lemma 2 gives $N_h(e_u, v) + B^*(e_p, v) = \mathcal{G}(v)$ on V_h^R , where $e_u := \tilde{u} - u_h$, $e_p := p^* - p_h$ and $p^* := \tilde{p} - \bar{p}_{\Gamma_h}(\tilde{p})$. Write $e_p = \eta + \xi$ with $\eta := p^* - \pi_h p^*$, π_h

the L^2 projection onto Π_h , and let $\bar{\xi}$ be the mean of ξ over Ω_h . Testing with V_h^0 and using the inf-sup condition gives $\beta\|\xi - \bar{\xi}\| \leq C(|e_u| + h_\Gamma^{3/2})$. Testing with $z - u_h \in Z_h$ leaves $\langle \eta + \xi - \bar{\xi}, (z - u_h) \cdot n \rangle$, and $|\langle \xi - \bar{\xi}, v \cdot n \rangle| \leq C(c_0 \gamma)^{-1/2} \|\xi - \bar{\xi}\| \|v\|$ by a discrete trace inequality; this is absorbed for $\gamma \geq \gamma_2$, which proves (b). With zero data the same argument gives $u_h = 0$ and $p_h \equiv c$, and an edge bubble with nonzero flux gives $c = 0$. On zero-flux test functions B^* is the naive form, and adding a constant turns a zero-flux solution into a CNS* solution; this proves (a). (c) follows with a Friedrichs inequality. $\square$

Remark 1 (Further results, proved in [6]). (1) *Pressure of GS* [6, Def. 8.10, Thm. 8.11]. For $G > 0$, fixed γ , bounded ρ and small h , $\min_{c' \in \mathbb{R}} \|p^* - p_h - c'\|$ and $\|p^* - p_h\|$ lie between $ch_\Gamma^{1/2} G/\gamma - C_p h_\Gamma$ and $Ch_\Gamma^{1/2} G/\gamma + C_p h_\Gamma$. (2) *CNS* uniformly in γ , and its pressure* [6, Thms. 7.30, 7.33, 8.7, Lemma 8.3]. With data $\tilde{g}_I$, Theorem 3(c) holds uniformly in $\gamma \geq \gamma_2$, and $\|p^* - p_h\|_{L^2(\Omega_h)} \leq C(h_\Gamma^{3/2} + h_\Omega^k)$ for $\gamma \leq \gamma_{\max}$. (3) *Sharpness*, every $k \geq 4$ [6, Thms. 7.8, 7.24]. Let $\theta_{\min}$ be the smallest angle of the triangles touching Γ_h , assume (S) $h_\Gamma \leq 2 \sin(\theta_{\min}/4)$, and let h_Γ be small. If $W = \partial_n u \cdot \tau \neq 0$ on Γ , then $\|\nabla(\tilde{u} - u_h)\| \geq c\|W\|h_\Gamma^{3/2} - Ch_\Gamma^2$ for CNS*, and for GS with any G , with c independent of γ, ρ, p . *Sketch.* On $v \in Z_h$ vanishing on Γ_h with $\int_e \partial_n v = 0$, pressure and penalty drop out, and the chordal bump $-\frac{1}{2} s^2 W_e$ of $\tilde{u} \cdot \tau_e$ ($H^{1/2}$ -size $h_\Gamma^{3/2}$) pairs with the second moment of $\partial_n v \cdot \tau_e$. At each wall vertex z , the curl of an explicit C^1 piecewise- P_5 stream function on three triangles makes this moment positive, by an exact identity, if the angle of Ω_h at z is below $\pi + \theta_{\min}$, which (S) ensures. (4) *A growing penalty* [6, Thm. 7.30, Cor. 7.34, Thm. 7.35]. For GS, $\mu \gtrsim hh_\Gamma^{-3/2}$ restores the H^1 rate 3/2 (data $\tilde{g}_I$), but no penalty scaling gives an energy rate above 1.

Proposition 4 (Penalty threshold [6, §10]). (a) $\mu \geq 4\kappa C_I^2 \rho$ suffices (Lemma 1). (b) Let T have an edge $e \in \Gamma_h$ and not meet ∂R . Map it by a similarity to the triangle with base $[(0, 0), (1, 0)]$ and apex (a, b) , with barycentric coordinates λ_0 (apex) and λ_1, λ_2 . For $t \in \mathbb{R}^3$ let $v_t := \text{curl}(\lambda_1^2 \lambda_2^2 (t_0 \lambda_0 + t_1 \lambda_1 + t_2 \lambda_2)) \in P_4^2$. Let A_T, B_T, C_T be the quadratic forms $\int_T \frac{1}{2} |Dv_t|^2$, $\int_e \partial_n v_t \cdot v_t$ and $\int_e |v_t|^2$, and let λ_T be the top eigenvalue of the pencil $(2B_T - A_T, C_T)$. If $\mu|e|/h < \lambda_T$, then N_h is indefinite on Z_h for every $k \geq 4$. (c) $\lambda_T > 1$ if the apex angle is at least 35° and the base angles at least 5° ; $\lambda_T > 4$ and $\lambda_T > 9.3$ for apex angles at least 45° and 60° . So if the triangle on a shortest edge of Γ_h has such a shape, coercivity on Z_h requires $\mu > \rho$.

Proof. (b) The potential vanishes to second order on the edges through the apex, so v_t , extended by zero, lies in Z_h . A_T and B_T are invariant under dilation and C_T scales with $|e|$, so $N_h(v_t, v_t) = (\mu|e|/h - \lambda_T)C_T(t) < 0$ for the top eigenvector. (c) A_T, B_T, C_T are explicit rational functions of (a, b) [6, §10.1]. On dyadic rational boxes covering each shape region we verify $2520 b^3 q^T (2B_T - A_T - \lambda C_T) q > 0$ for a rational q in exact interval arithmetic (297 boxes for $\lambda = 1$) [6, Prop. 10.3]. For tall triangles $\lambda_T < 0$, and even one-star witnesses can fail [6, §10]; necessity under (M0)–(M2) alone is open. $\square$

Table 1: H^1 seminorm errors (rate), tests A and B₁₀₀, and pressure errors $\|p^* - p_h\|_{L^2(\Omega_h)}$ for B₁₀₀. Strong: $u_h = 0$ at the nodes of Γ_h , mesh (a) (alternating first-layer diagonals; wall vertices in two or four triangles, violating (M2)). GS, CNS*: standard mesh. Rates from the preceding $N \in \{16, 24, 32, 48, 64, 96, 128, 192\}$; at $N = 256$, GS and CNS* give 1.55, 1.56 on test A.

N	test A, velocity			test B ₁₀₀ , velocity		test B ₁₀₀ , pressure	
	strong (a)	GS	CNS*	GS	CNS*	GS	CNS*
32	5.72e-1 (0.72)	5.55e-2 (1.79)	6.41e-2 (1.80)	3.51 (0.93)	2.34e-2 (2.43)	18.2 (0.68)	7.67e-2 (3.03)
64	3.75e-1 (0.62)	1.81e-2 (1.68)	2.08e-2 (1.69)	1.88 (0.97)	6.83e-3 (1.76)	11.8 (0.65)	1.62e-2 (2.08)
96	2.98e-1 (0.59)	9.57e-3 (1.63)	1.09e-2 (1.64)	1.28 (0.98)	3.54e-3 (1.68)	9.23 (0.62)	8.01e-3 (1.80)
128	2.54e-1 (0.56)	6.11e-3 (1.59)	6.96e-3 (1.60)	9.71e-1 (0.98)	2.24e-3 (1.63)	–	–
192	–	3.27e-3 (1.57)	3.71e-3 (1.58)	6.54e-1 (0.99)	1.18e-3 (1.60)	–	–

4. Numerical study

We take $k = 4$, $L = 2.5$, and N polygon vertices at the radial projections of N equispaced points on the square; then $h/h_\Gamma \approx 3.3$, $\max |e|/\min |e| \leq 1.97$, and each vertex of Γ_h lies in three triangles. Direct solves give relative residuals below 10^{-12} and $\|\operatorname{div} u_h\| \leq 2 \cdot 10^{-9}$ (10^{-6} for strong, mesh (a)). We use $\mu = 100$ ($\gamma \approx 30$), known to be above the measured coercivity threshold only. Test A is Scott’s benchmark $u = (1 - r^{-2})(-y, x)$, $p = 0$. Test B_a has stream function $(r^2 - 1)^2(x + y^2/2)/10$ and $p = a(x^2y - y + x/2)$, so $G = a(7\pi/8)^{1/2}$.

In Table 1 strong imposition locks (rate $\rightarrow 1/2$), while on test A ($G = 0$) both Nitsche methods tend to $3/2$. On our standard mesh, which satisfies (M2), strong imposition does not lock (rate 1.61 at $N = 128$). On B₁₀₀ the printed method has H^1 rate 1, an error over 500 times that of CNS* at $N = 192$, pressure rate decreasing towards $1/2$, and a maximum pressure error that does not decrease (77 at $N = 96$). The CNS* rates decrease monotonically towards $3/2$; its errors for B₁ and B₁₀₀ agree to all printed digits. On the pure-pressure test ($u = 0$, $f = \nabla p$, p as in B₁), CNS* gives $u_h = 0$ to round-off; at $N = 96$ the normalised slip and energy errors of Theorem 1(c) are 0.981 and 0.996 for $\mu = 10^2$, and 0.9993, 0.9994 for $\mu = 10^4$, outside that regime.

We locate μ^* by bisection on the inertia of the velocity block augmented by a divergence penalty. With Γ_h fixed and the far field coarsened by enlarging the box ($L = 2.5, 5, 10, 20, 40$), μ^* is linear in ρ over more than a decade ($\mu^* = 59.3, 112, 208, 411, 809$ for $\rho = 4.70, 8.47, 16.2, 31.8, 63.0$ at $N = 16$; similarly at $N = 32$), so $\gamma^* = \mu^* h_\Gamma/h \approx 18$ – 21 and $\mu^*/\rho \approx 11$ – 13 ; under refinement at $L = 2.5$, $\gamma^* = 14.7, 18.1, 19.9, 20.8$ for $N = 8, 16, 32, 64$. This sweep fixes $\max |e|/\min |e|$, so it cannot separate ρ from h/h_Γ . The certified $\lambda_T > 9.3$ is a local lower bound; the gap is due to the flatter first layer and collective boundary modes [6, §10.1]. On these meshes ($k = 4$), $\mu \geq 25\rho$, which implies $\gamma \geq 25 > \gamma^*$, was always coercive, while a fixed μ fails once $h/h_\Gamma \gtrsim \mu/20$.

CRedit authorship contribution statement

Claudemir de Souza Cavalcante: FILL. **Jaideep Sai Padhi:** Formal analysis, Software, Validation, Writing – original draft.

Declaration of competing interest

The question studied is the subject of a prize offered by L. R. Scott; the authors intend to submit this work for it. The authors declare no other competing interests.

Data availability

Code, raw outputs and logs for every number reported here are at <https://github.com/jaideepsaipadhi/zero-gradient-prize> and on Zenodo, doi:10.5281/zenodo.XXXXXXXX.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) in order to develop and check proofs, write and run code, verify the literature, and draft and edit the text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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